is reachable from `q`, and this goes against our initial hypothesis.

It follows that `w` can't be a core point: it must be a non-core points reachable from at least 2 different core points, a situation illustrated in figure 12.11.

In these cases, reachable points can be added to either cluster, but the difference will be just about a single point; it also means that the two clusters are separated by an area with lower-than-threshold density (although not completely empty).

The final result won't be influenced by the order these points are processed, because sooner or later we will discover that all of them belong to the same cluster. Nevertheless, there is an efficient way and a bad way to process the points: depending on the order we follow, we might be forced to use different methods to keep track of the clusters.

The bad way is: if we processed points in a completely random order, then we would need to keep track of the cluster assigned to each point (initially each point being in its own cluster…), but we would also need to keep track of which clusters need to be merged (every time we process a core point, we'll have to try[15] to merge at least `minPoints-1` clusters); this becomes complicated to handle and requires an ad-hoc data structure, the *disjoint set* we described in chapter 5.